🧀 🧀 12 Meeting at Kinopark

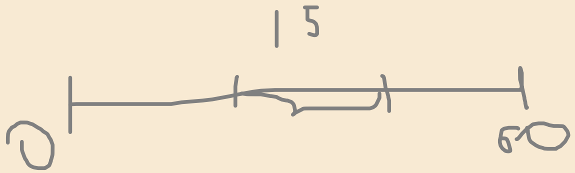
Anush and Nairi are shopping at the mall. They agree to split up for a time and then meet for lunch. They plan to meet in front of Kinopark between 12:00 and 13:00. The one who arrives first agrees to wait 15 minutes for the other to arrive. After 15 minutes, that person will leave and continue shopping. What is the probability that they will meet if each one of them arrives at any time between 12:00 and 13:00?

Hint: Represent on the coordinate plane with x = Anush's arrival, y = Nairi's arrival.

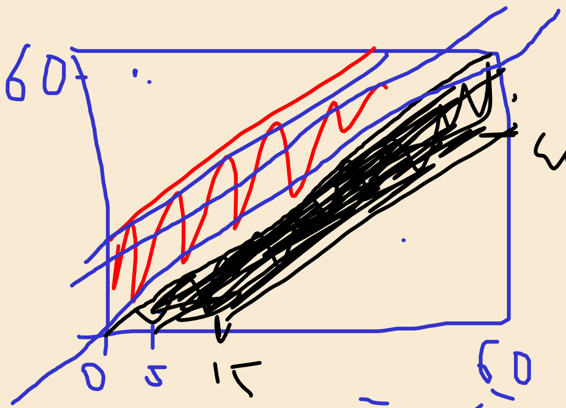
12
13
16
17
18

10 5

10 friends



$|x-y| \le 15$

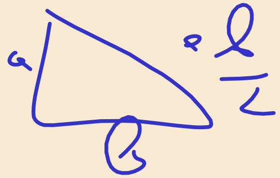


$y = x$

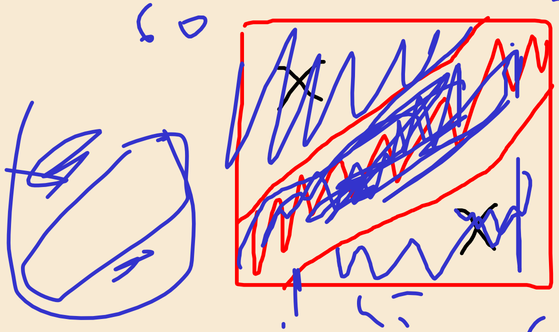
$y = x-2$

$y \geq x-15$

$y \leq x+15$



$60^2 - 45$



$\frac{45 \cdot 45}{2}$

$\frac{\#}{\# \text{total}}$

$60^2 - \frac{45^2 \cdot 2}{2}$

3600

🧀 🧀 16 Commute Modes and Lateness

Nune uses her car 30% of the time, walks 30% of the time, and rides the bus 40% of the time to work. She is late 10% of the time when she walks, 3% of the time when she drives, and 7% of the time when she takes the bus.

- a. Yesterday she was late. What is the probability she took the bus?
- b. Today she was on time. Do you think she walked?

N → 30, 30, 40

$P(L) =$

$0.3 \cdot 0.03 +$

$0.3 \cdot 0.1 +$

$0.4 \cdot 0.07$

$0.009 +$

$0.03 +$

0.028

0.067

$= \frac{28}{67}$

$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$

B - bus L - Late

$P(B|L) = \frac{P(L|B) \cdot P(B)}{P(L)} = \frac{0.07 \cdot 0.4}{0.067}$

$P(B \cap L) = P(B|L)\, P(L) = P(L|B) \cdot P(B)$

$$P(L^c) = 1 - 0.067$$

$$P(W \mid L^c) = \frac{P(L^c \mid W) \cdot P(W)}{P(L^c)}$$

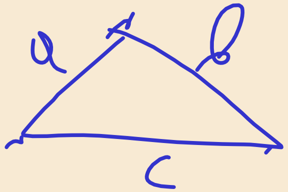
$$= \frac{0.9 \cdot 0.3}{0.933} = \frac{270}{933}$$

$$\approx 0.3$$

$a, b, c \in U[0,1]$

q. 3

13

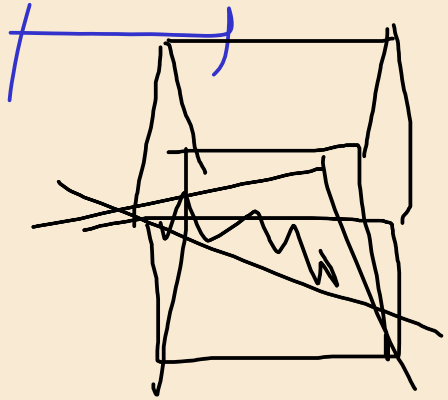


$\boxed{a < b + c}$

$b < a + c$

$c < a + b$

$\boxed{3 \cdot P(c > a + b)}$



🧀 🧀 17 Two-Headed Coin Bayes

Rosie has ten coins. Nine of them are ordinary coins with equal chances of Heads and Tails when tossed, and the tenth has two Heads.

- a. If she takes one of the coins at random from her pocket, what is the probability that it is the coin with two Heads?
- b. If she tosses the coin and it comes up Heads, what is the probability that it is the coin with two Heads?
- c. If she tosses the coin one further time and it comes up Tails, what is the probability that it is one of the nine ordinary coins?

HT HT HT

11 H

9 T

a) $\boxed{P(2\text{-h}) = \frac{1}{10}}$

b) $P(2\text{-h} \mid H)$

$\frac{2}{11}$

$= \frac{1}{10} + \frac{9}{20} = \frac{11}{20}$

$P(H \mid 2\text{-h}) = 1$

$P(H \mid 2\text{-h}^c) = \frac{1}{2}$

$P(H) = P(2) \cdot P(H \mid 2) + P(2^c) \cdot P(H \mid 2^c)$

$\frac{1}{10} \cdot 1 \qquad \frac{9}{10} \quad \frac{1}{2}$

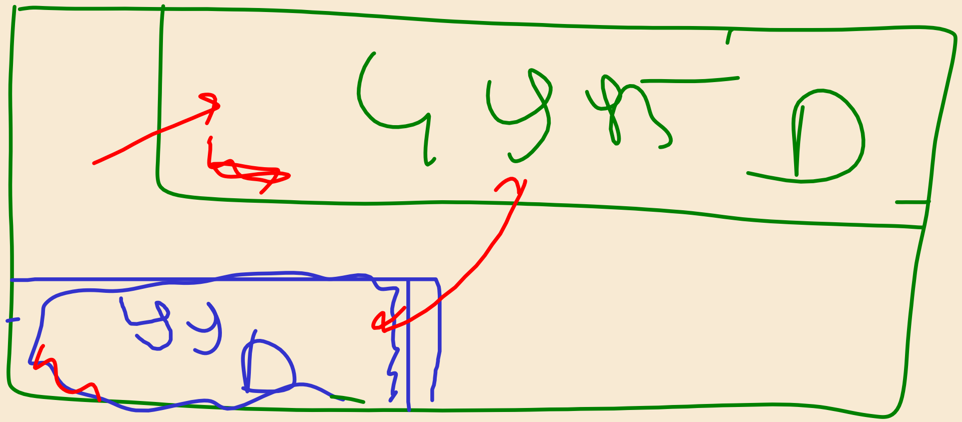
$P(2 \mid H) = \frac{P(H \mid 2)\, P(2)}{P(H)} = \frac{1 \cdot \frac{1}{10}}{\frac{11}{20}} = \frac{2}{11}$

$$P(D) = \boxed{0.001}$$

$$P(+ \mid D) = 0.99$$

$$P(\pm \mid D^c) = 0.05$$

$$P(D \mid +)$$



$$\frac{99}{4995 + 99} = \boxed{2\%}$$

$\boxed{100k}$

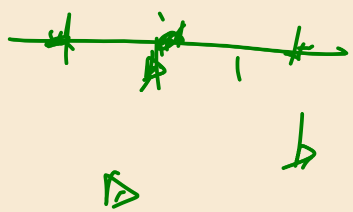
99.9k − [illegible]

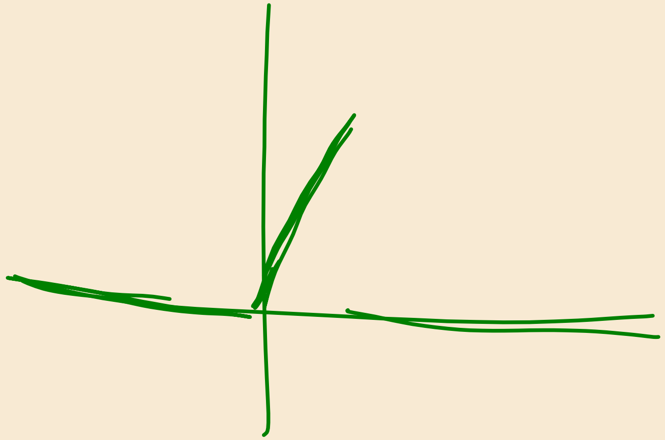
$$99.900 \cdot 0.05 = 4995$$

$$\boxed{\frac{1}{1000} \cdot \frac{99}{100}} \ll 99.9\% \cdot 0.05$$

X

$$f_X(x) = \begin{cases} 2x, & x \in [0,1] \\ 0 \end{cases}$$





$1, 2, 3, 4, 5, 6$

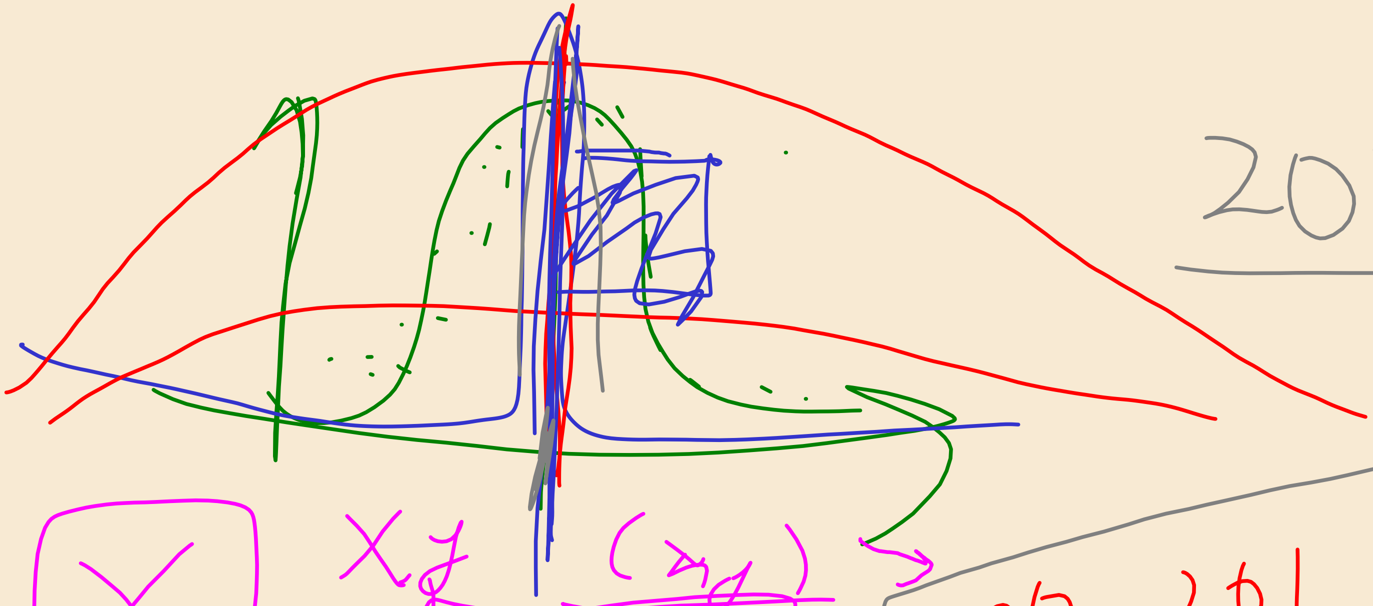
$\frac{1}{6}\cdot 1 + \frac{1}{6}\cdot 2 + \frac{1}{6}\cdot$

$\sum x p(x)$

$$\mathbb{E}(X) = \int_{-\infty}^{0} x \cdot 0 \, dx + \int_0^1 x \cdot 2x \, dx +$$

$$\int_0^1 2x^2 \, dx$$

$= 0$



200A

200 201 199

200

10000

101010

5.56-1y

$E(X - E(X))^2$

$E(X)^2 - E(X^2)$

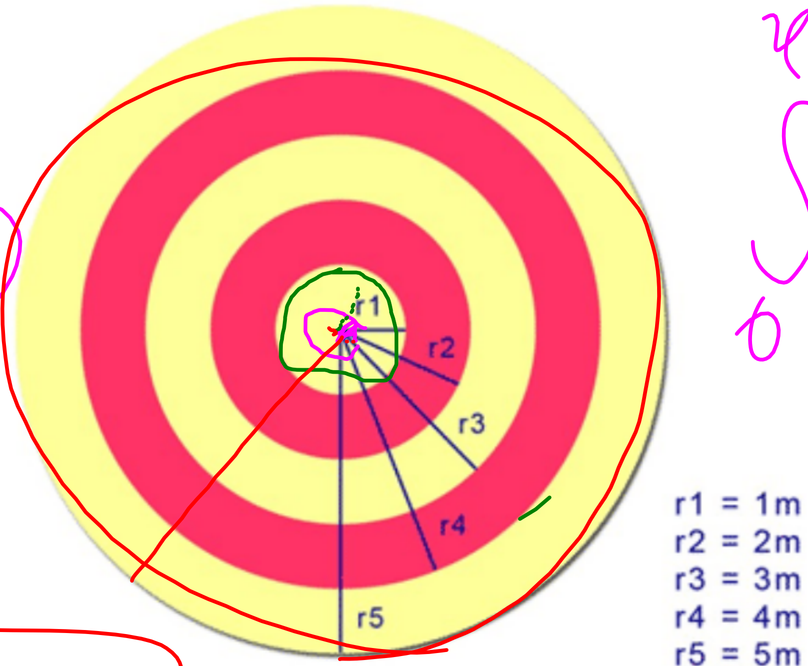
$E X^2$

A dart is thrown at a circular target with concentric circles. Circle 1 (innermost) has radius 1m, and each subsequent radius increases by 1m. Find the probability that the dart lands in:

a. circle 1
b. a red circle
c. a yellow circle

$$f(x) = \frac{1}{x}$$

$$\int_0^{r_1} \frac{1}{r} \frac{\pi r^2}{2}$$



dart_throwing

$$\frac{1}{2\sqrt{2\pi}\sigma} e^{-x}$$

$$\int_0^{r_1} f(x)$$

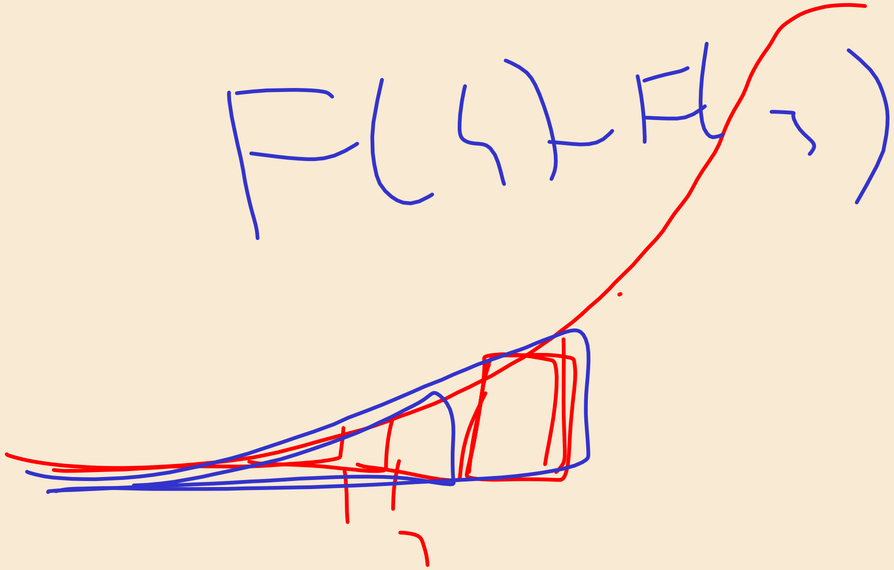
$$F(X < r_1)$$

$$F_1(r_1) = P(x < r_1)$$

$$= \int_0^{r_1} x f(x)\,dx$$

F(4)+F(-1)

X+X



1	7
5	3.5+3.5
3	2F(X)

1.66-1

.66

3(6-1)

1	5
2	4
4	2
3	3
5	1

$\boxed{X+X}$

6 Unit

X+

5

f(x)

36

yx

$\boxed{\sum f(x)\, g(y-x)}$

1 - 1/6

2 - 1/6

3 - 1/6